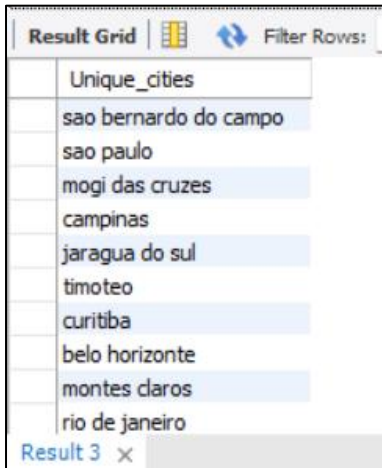


Project Summary

Project Overview : This project provides a 360-degree analysis of an E-commerce marketplace using SQL and Python. The objective is to extract business-critical metrics from raw data and visualize trends to help in data-driven decision-making.

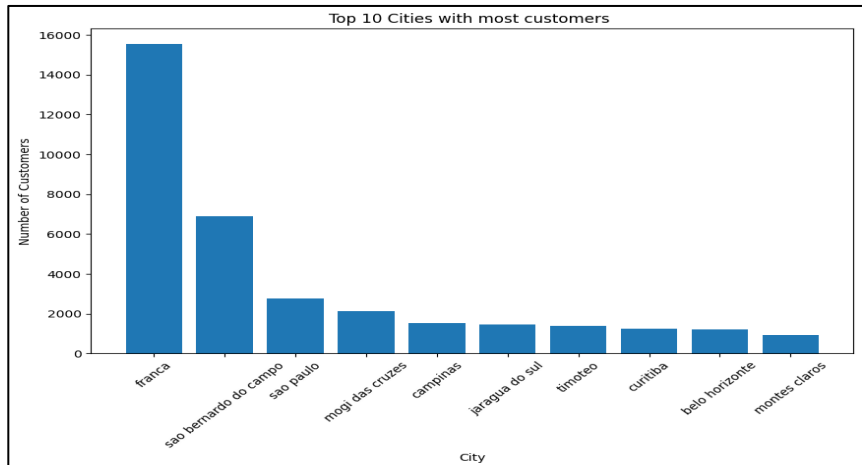
Analysis and visualizations :

1. List all unique cities where customers are located.



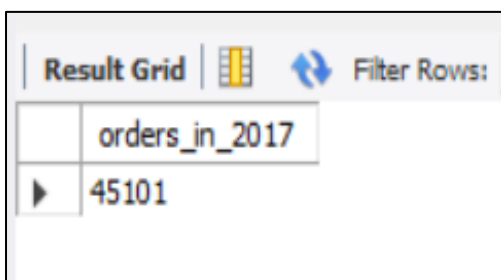
The screenshot shows a Jupyter Notebook interface with a 'Result Grid' tab. It displays a list of unique cities under the column header 'Unique_cities'. The cities listed are: sao bernardo do campo, sao paulo, mogi das cruces, campinas, jaragua do sul, timoteo, curitiba, belo horizonte, montes claros, and rio de janeiro. The interface also includes a 'Filter Rows' button and a 'Result 3' indicator.

Unique_cities
sao bernardo do campo
sao paulo
mogi das cruces
campinas
jaragua do sul
timoteo
curitiba
belo horizonte
montes claros
rio de janeiro



Insight: Most of our customers live in Sao Paulo. We should focus our marketing and fast delivery services in this city.

2. Count the number of orders placed in 2017.



The screenshot shows a Jupyter Notebook interface with a 'Result Grid' tab. It displays the count of orders in 2017 under the column header 'orders_in_2017'. The value shown is 45101. The interface also includes a 'Filter Rows' button.

orders_in_2017
45101

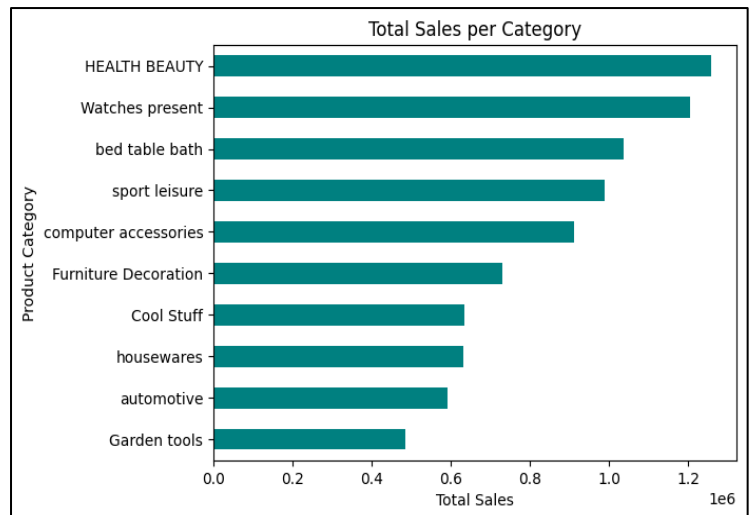
```
# number of orders placed in 2017
order_2017= order[order['order_purchase_timestamp'].dt.year == 2017]
print("Total order placed in 2017:", len(order_2017))
```

Total order placed in 2017: 45101

Insight: We received 45,101 orders in 2017. We can use this number to set our sales targets for next year.

3. Find the total sales per category.

Result Grid			Filter Rows:
	total_sales	product_category	
▶	1258681.34	HEALTH BEAUTY	
	1205005.68	Watches present	
	1036988.68	bed table bath	
	988048.97	sport leisure	
	911954.32	computer accessories	
	729762.49	Furniture Decoration	
	635290.85	Cool Stuff	
	632248.66	housewares	
	592720.11	automotive	
	485256.46	Garden tools	

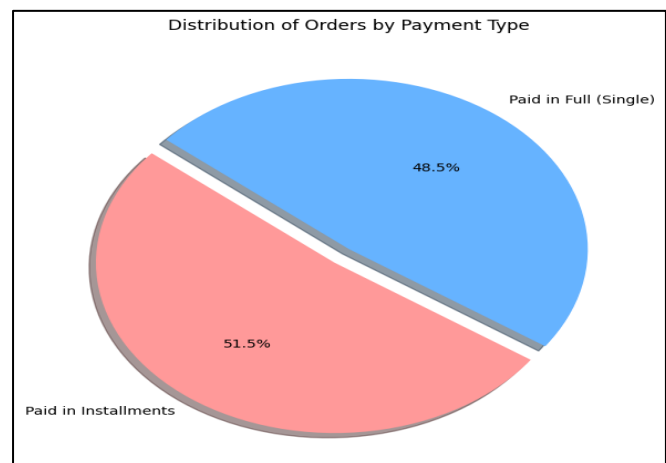


Insight: "Health Beauty" and "Watches" generate the highest sales. We should promote these items more on the homepage.

4. Calculate the percentage of orders that were paid in installments.

Result Grid			Filter Rows:
	total_orders	paid_in_installments	
▶	99440	51.63	

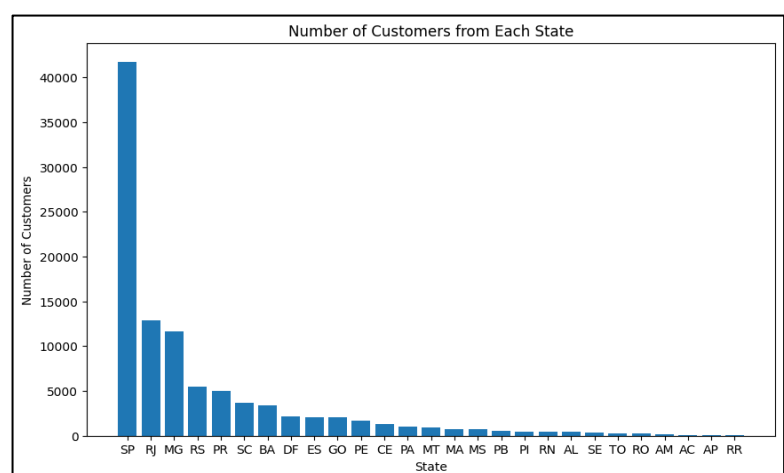
Result 3 x



Insight: Around 51% of customers choose to pay in installments. Keeping this payment option is extremely important to avoid losing sales.

5. Count the number of customers from each state

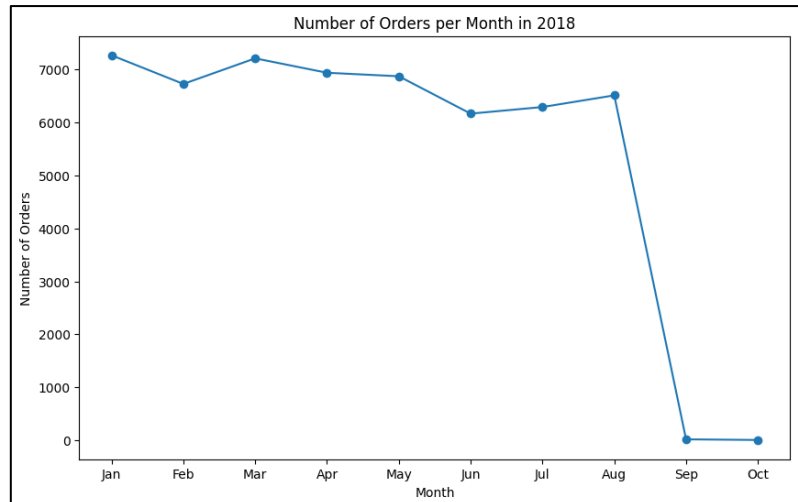
Result Grid			Filter Rows:
	count(customer_id)	customer_state	
▶	41746	SP	
	3637	SC	
	11635	MG	
	5045	PR	
	12852	RJ	
	5466	RS	
	975	PA	
	2020	GO	
	2033	ES	
	3380	BA	
	747	MA	
	715	MS	
	1336	CE	



Insight: The state of Sao Paulo has the most customers by a huge margin. We are very strong in this region compared to others.

6. Calculate the number orders per month in 2018.

Result Grid			 Filter Rows
	order_month	total_order	
▶	1	7269	
	2	6728	
	3	7211	
	4	6939	
	5	6873	
	6	6167	
	7	6292	
	8	6512	
	9	16	
	10	4	



Insight: Orders were very steady (around 6,000 to 7,000 a month) until August. The sudden drop in September and October is likely just incomplete data, not a real drop in sales.

7. Find the average number of products per order, grouped by customer city.

Result Grid		 Filter Rows:
customer_city	avg_products_per_order	
padre carvalho	7.0000	
celso ramos	6.5000	
candido godoi	6.0000	
datas	6.0000	
matias olimpio	5.0000	
cidelandia	4.0000	
curralinho	4.0000	
morro de sao paulo	4.0000	
picarra	4.0000	

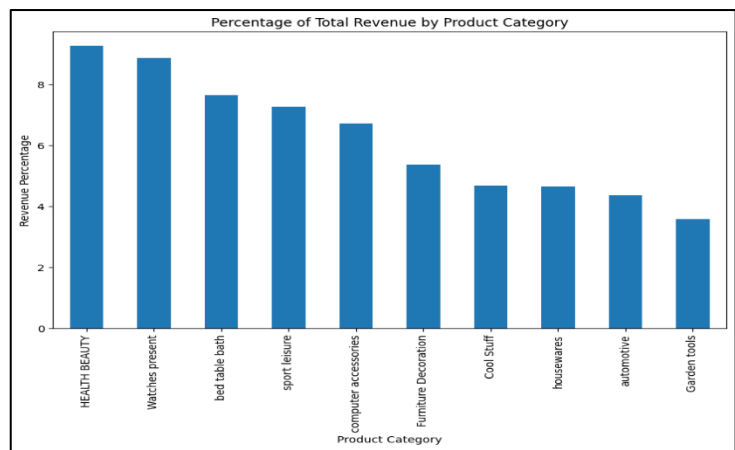
```
customer_city
padre carvalho      7.0
celso ramos         6.5
datas               6.0
candido godoi       6.0
matias olimpio      5.0
morro de sao paulo  4.0
picarra             4.0
curralinho          4.0
cidelandia          4.0
teixeira soares     4.0
Name: products_count, dtype: float64
```

Insight: Customers in some smaller cities buy many items at once (an average of 7 items per order). These might be wholesale buyers or small businesses.

8. Calculate the percentage of total revenue contributed by each product category.

Result Grid			Filter Rows:
	product_category	total_revenue	
▶	HEALTH BEAUTY	9.26	
	Watches present	8.87	
	bed table bath	7.63	
	sport leisure	7.27	
	computer accessories	6.71	
	Furniture Decoration	5.37	
	Cool Stuff	4.67	
	housewares	4.65	
	automotive	4.36	

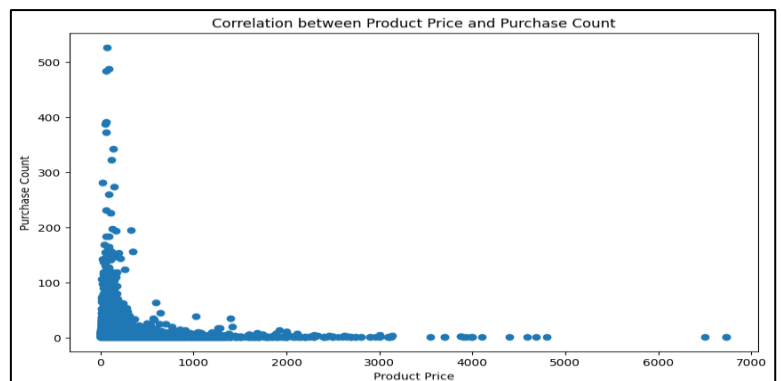
9.



Insight: Our sales are nicely balanced across many categories. The top category only makes up 9.26% of revenue, meaning we aren't completely dependent on just one type of product.

9. Identify the correlation between product price and the number of times a product has been purchased.

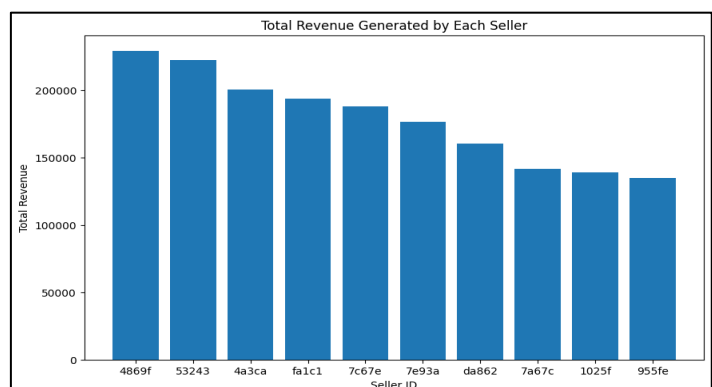
Result Grid			Filter Rows:
	correlation_coefficient		
▶	-0.032139862547783636		



Insight: There is no connection between an item's price and how often it is bought. Cheaper items do not automatically sell faster than expensive ones.

10. Calculate the total revenue generated by each seller, and rank them by revenue.

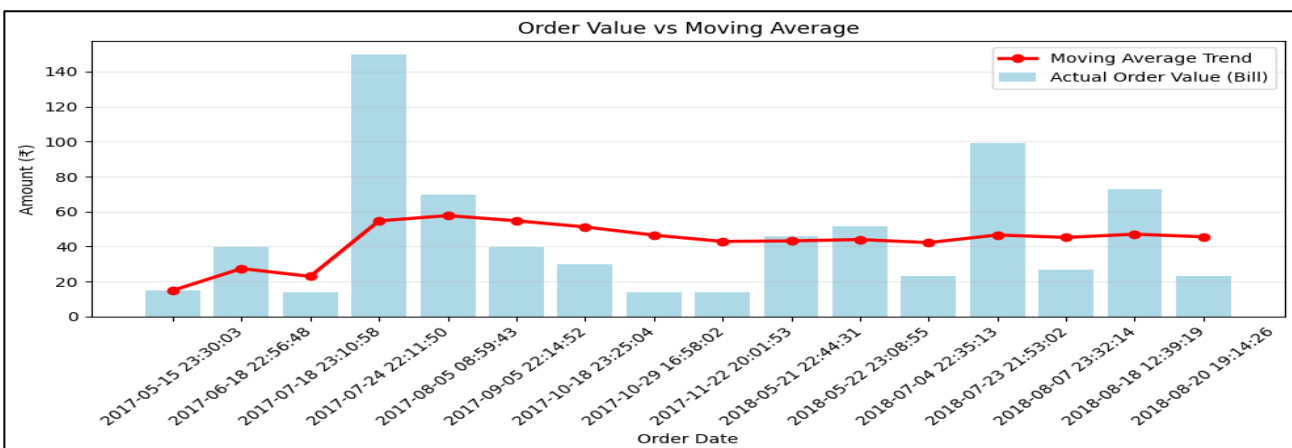
Result Grid			Filter Rows:	Exp
	total_revenue	seller		
▶	229472.63	4869f7a5dfa277a7dca6462dcf3b52b2		
	222776.05	53243585a1d6dc2643021fd1853d8905		
	200472.92	4a3ca9315b744ce9f8e9374361493884		
	194042.03	fa1c13f2614d7b5c4749cbc52fecda94		
	187923.89	7c67e1448b00f6e969d365cea6b010ab		
	176421.07	7a02a12af20c4f02f29b302470b77c2a		



Insight: A few top sellers are bringing in the most money. We need to make sure these top sellers get good support so they don't leave our platform.

11. Calculate the moving average of order values for each customer over their order history.

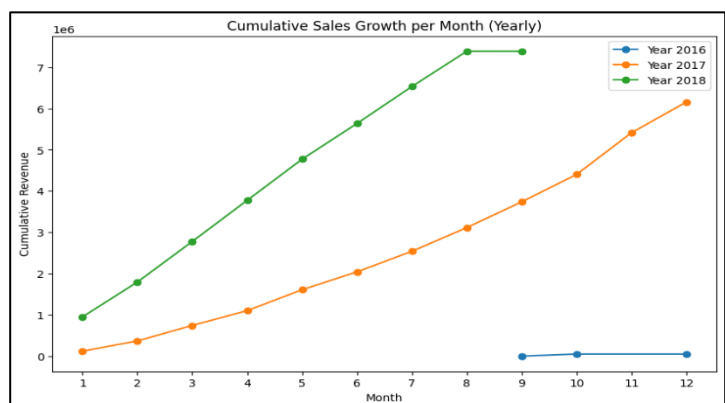
customer_unique_id	order_purchase_timestamp	order_value	moving_avg
0000366f3b9a7992bf8c76cdf3221e2	2018-05-10 10:56:27	129.89999389648438	129.89999389648438
0000366f3b9a7992bf8c76cdf3221e2	2018-05-10 10:56:27	129.89999389648438	129.89999389648438
0000b849f77a49e4a4ce2b2a4ca5be3f	2018-05-07 11:11:27	18.899999618530273	18.899999618530273
0000b849f77a49e4a4ce2b2a4ca5be3f	2018-05-07 11:11:27	18.899999618530273	18.899999618530273
0000f46a3911fa3c0805444483337064	2017-03-10 21:05:03	69	69
0000f46a3911fa3c0805444483337064	2017-03-10 21:05:03	69	69



Insight : Tracking the moving average helps us see the normal spending habits of customers, ignoring sudden spikes from random large orders.

12. Calculate the cumulative sales per month for each year.

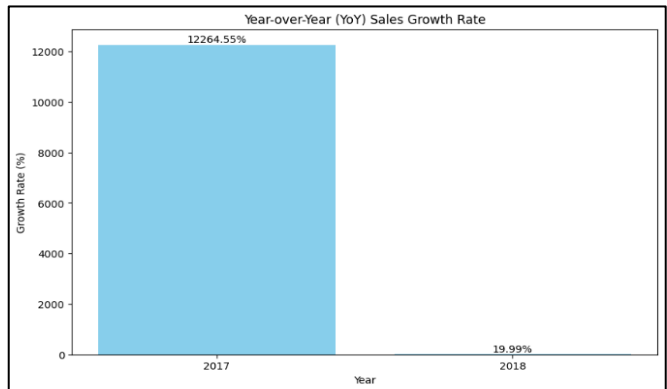
month	year	monthly_sales	cumulative_sales
9	2016	534.7200164794922	534.7200164794922
10	2016	99015.31955718994	99550.03957366943
12	2016	21.799999237060547	99571.8395729065
1	2017	240625.73960590363	240625.73960590363
2	2017	494606.03963184357	735231.7792377472
3	2017	748688.598991394	1483920.3782291412
4	2017	719854.4594259262	2203774.8376550674
5	2017	1012142.2786035538	3215917.116258621



Insight: Sales grew steadily month after month in 2017. This shows healthy, continuous growth without any slow periods.

13. Calculate the year-over-year growth rate of total sales.

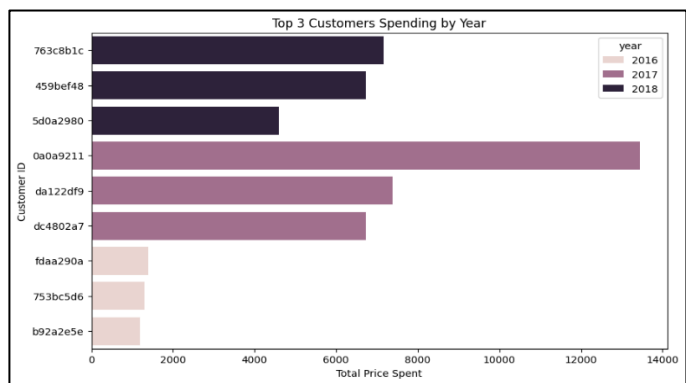
year	revenue	lastyear_revenue	yoy_growth_rate
2016	99571.8395729065	NULL	NULL
2017	12311613.959013939	99571.8395729065	12264.55
2018	14772101.604853868	12311613.959013939	19.99



Insight: The massive 12,264% growth in 2017 likely means we only have a tiny bit of data for 2016. The 20% growth in 2018 is a much more realistic picture of how the business is doing.

14. Identify the top 3 customers who spent the most money in each year.

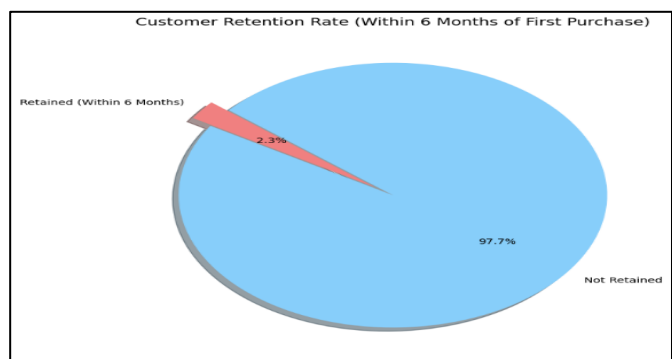
order_year	customer_id	total_spent
2018	ec5b2ba62e574342386871631fafd3fc	14320
2018	f48d464a0baaea338cb25f816991ab1f	13458
2018	e0a2412720e9ea4f26c1ac985f6a7358	9199.7998046875
2017	1617b1357756262bfa56ab541c47bc16	26880
2017	c6e2731c5b391845f6800c97401a43a9	13470
2017	3fd6777bbce08a352fddd04e4a7cc8f6	12998
2016	a9dc96b027d1252bbac0a9b72d837fc6	2798
2016	1d34ed25963d5aae4cf3d7f3a4cda173	2599.97998046875
2016	4a06381959b6670756de02e07b83815f	2398



Insight: A few key customers spend a huge amount of money. We should offer them VIP discounts to thank them and keep them loyal.

15. Calculate the retention rate of customers, defined as the percentage of customers who make another purchase within 6 months of their first purchase.

total_customers	retained_customers	retention_rate
96096	2234	2.32



Insight: Only 2.3% of customers come back to buy again within 6 months. This is very low. We need to send better emails and offers to get people to shop a second time.

Key Business Recommendations:

Launch a Customer Retention Campaign: Since only 2.3% of customers return within 6 months, we urgently need to create an automated email campaign offering a special discount for a customer's second purchase.

Focus Marketing on Sao Paulo: The data clearly shows Sao Paulo is our strongest market. We should allocate the largest portion of our advertising budget and offer faster shipping options in this specific region to maximize sales.

Promote Installment Options: With over 50% of orders paid in installments, this feature is critical to our success. We should make "Pay in Installments" highly visible on every product page to encourage more people to buy.

Create a VIP Seller Program: A small group of top sellers generates most of the revenue. We should assign them dedicated support managers or give them lower platform fees to ensure they don't leave for a competitor.